



Draw a Square with a Loop

Quick facilitation notes & answer key

Goal

Students use a for-loop to draw a square: repeat 4 of forward + right 90. They see the forward number sets size while `range(4)/right 90` keep it a square, and link the path length to perimeter.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

Python Turtle projected (or a web playground); the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Recall last sheet: `range(3)` drew 3 sides — add the 4th, it closes a square.
2. Run the `range(4)` square; note it ends back at the start.
3. Students do Exercises 1–4 (Ex 4 links to perimeter), solo or in pairs.
4. Share answers; land the big idea: `forward = size`; `range(4)/right 90 = square`.

Answer key (checked by the run-gate)

Ex 1 — 4 sides; 4 turns.

Ex 2 — still a square; the SIZE changes, not the shape (perimeter 400).

Ex 3 — `range(4)`, `bit.right(90)`.

Ex 4 (challenge) — 200 (4×50) — the perimeter.

Watch for & differentiate

Common slip: changing `range(4)` when only the size should change (Ex 2).

Support: trace one side and skip-count 50 four times for the perimeter.

Extend: ask the perimeter of a `forward(70)` square (280).

Success looks like

The student reads the square loop (Ex 1), reasons size-vs-shape when altering forward (Ex 2), completes the loop (Ex 3), and finds the perimeter from the code (Ex 4).